

SYSTEM CAPACITY & CARE LOAD ANALYTICS FOR UNACCOMPANIED CHILDREN

A DATA-DRIVEN FRAMEWORK FOR MONITORING CARE LOAD, CAPACITY PRESSURE, AND SYSTEM SUSTAINABILITY

Abstract

The Unaccompanied Alien Children (UAC) Program operates as a dynamic care pipeline in which children apprehended by U.S. Customs and Border Protection (CBP) may move through CBP custody, transfer into the care of the U.S. Department of Health and Human Services (HHS), receive medical, psychological, and welfare support, and eventually be discharged through placement with sponsors.

This project develops a data-driven analytical framework for monitoring the capacity and care load of this system. Daily operational data covering the period from January 12, 2023 through December 21, 2025 were analyzed. The dataset contained 720 reporting periods and six primary operational variables covering apprehensions, CBP custody, transfers from CBP, HHS care, and HHS discharges.

Derived indicators were created to measure total system load, net intake pressure, care-load growth, rolling flow averages, backlog accumulation, and capacity pressure. Additional analysis classified reporting periods according to system-load thresholds, identified prolonged high-load windows, identified capacity-relief periods, compared yearly system behavior, and generated a system-load forecast.

The analysis identified a peak system load of 11,762 children on December 20, 2023. Two prolonged high-load windows were identified: August 22–October 19, 2023 and November 21, 2023–January 4, 2024. The yearly analysis showed a substantial decline in average system load from 8,846.54 children in 2023 to 7,319.86 in 2024 and 2,575.74 in 2025. These findings demonstrate the value of continuous capacity monitoring and provide an analytical foundation for staffing, shelter planning, operational preparedness, and humanitarian response evaluation.

Keywords: UAC, healthcare analytics, capacity analytics, care load, CBP, HHS, time-series analysis, data science, operational analytics, forecasting

1. INTRODUCTION

The UAC care system is a complex operational pipeline involving multiple stages of custody, transfer, care, and discharge. Effective management of this system requires awareness of both the number of children currently receiving care and the rate at which children enter and leave different stages of the pipeline.

Sudden increases in arrivals can increase pressure on CBP custody and HHS care facilities. Conversely, sustained discharge activity can reduce system inventory and create periods of operational relief. Monitoring only individual daily counts may therefore provide an incomplete picture of system conditions.

This project applies data analytics techniques to transform daily operational data into a structured capacity-monitoring framework. The analysis focuses on identifying changes in total system load, inflow and outflow balance, periods of elevated pressure, prolonged high-load windows, relief periods, and longer-term changes in system behavior.

The project also provides a Streamlit-based interactive dashboard so that the analytical findings can be explored through visualizations, KPI summaries, date filters, and forecast information.

2. BACKGROUND AND CONTEXT

The UAC Program involves children who enter federal custody and subsequently move through different stages of care.

From a healthcare-systems perspective, the process can be represented as a dynamic pipeline:

CBP intake → CBP custody → Transfer to HHS → HHS care → Discharge/placement

Each stage represents an operational responsibility and requires appropriate resources.

CBP custody represents an initial care load, while HHS care represents children receiving services after transfer. The HHS stage may involve medical screening, quarantine, sheltering, psychological and welfare support, and preparation for placement.

The overall system therefore behaves like a dynamic inventory system. The number of children under care at a given time depends on the balance between incoming children, transfers between stages, and eventual discharge.

This makes capacity analytics particularly useful because a high inventory may persist even when current inflow is declining, while a lower inventory may still be associated with increasing pressure if inflow begins to exceed outflow.

3. PROBLEM STATEMENT

Although daily operational information is available, individual operational variables do not by themselves provide a centralized view of system capacity.

Without structured analytics, it can be difficult to determine:

- How many children are under care across the complete system.
- Whether system inventory is increasing or decreasing.
- When the system experiences sustained high-load conditions.
- When relief periods occur.
- Whether pressure is concentrated within particular periods.
- How system behavior changes from year to year.
- Whether historical trends indicate potential future changes in system load.

A centralized analytical framework can help transform these individual measures into operational indicators that are easier to interpret.

4. OBJECTIVES

4.1 Primary Objectives

The primary objectives of this project were to:

1. Quantify daily and cumulative care load across CBP and HHS.
2. Measure the balance between inflow and outflow.
3. Identify periods of capacity pressure and relief.
4. Detect sustained high-load periods.
5. Analyze changes in system load over time.

4.2 Secondary Objectives

The project also aimed to:

- Support healthcare staffing and shelter planning.
- Improve operational situational awareness.
- Provide data-driven evidence for capacity monitoring.
- Support humanitarian response evaluation.
- Provide an interactive dashboard for decision support.

- Develop a forecast of future system load using the completed analytical dataset.

5. DATASET DESCRIPTION

The analysis dataset consisted of daily UAC operational records from:

January 12, 2023 to December 21, 2025

The final prepared dataset contained:

720 reporting periods and 6 primary operational variables.

The principal variables were:

Variable	Description
Date	Reporting date
Children apprehended and placed in CBP custody	Daily intake volume
Children in CBP custody	Active CBP care load
Children transferred out of CBP custody	Flow from CBP toward HHS
Children in HHS care	Active HHS care load
Children discharged from HHS care	HHS outflow associated with placement/discharge

The data were converted to a standardized daily time-series structure, dates were converted to datetime format, and the observations were arranged chronologically.

6. DATA PREPARATION

The data-preparation stage included:

- Loading the operational dataset.
- Standardizing column names.
- Converting the date field to datetime format.
- Sorting records chronologically.
- Reviewing the structure of the dataset.
- Checking the reporting period.
- Preparing the dataset for derived metric calculations.

The final preparation output confirmed:

Date range: January 12, 2023 – December 21, 2025

Shape: 720 rows × 6 columns

The prepared dataset formed the foundation for all subsequent capacity analysis.

7. DERIVED CAPACITY METRICS

Several analytical variables were generated from the operational data.

7.1 Total System Load

Total system load represents the combined number of children in CBP custody and HHS care.

It provides a system-wide view of the children currently under federal care within the two monitored stages.

7.2 Net Intake Pressure

Net intake pressure was used as a flow-balance indicator comparing relevant HHS inflow and discharge/outflow measures.

A positive value indicates greater measured inflow relative to outflow for the corresponding period, while a negative value indicates that measured outflow exceeded the corresponding inflow.

This metric should be interpreted as an operational flow indicator rather than a causal measure.

7.3 Care Load Growth Rate

The care-load growth rate measures the percentage change in total system load from one reporting period to the next.

This allows periods of rapid increase or decrease to be identified.

7.4 Rolling Averages

Seven-period and fourteen-period rolling averages were calculated for relevant system and flow metrics.

Rolling averages reduce the influence of short-term fluctuations and help identify sustained changes in operational conditions.

7.5 Backlog Accumulation Rate

The backlog accumulation indicator was designed to identify sustained positive pressure in the care system.

It provides additional context beyond a single-day flow measurement.

7.6 Capacity Pressure

Capacity pressure was classified using the project's derived analytical conditions.

The initial analysis identified:

Capacity classification	Number of periods
Relief	468
Moderate Pressure	205
High Pressure	45
Balanced	2

A refined capacity analysis subsequently identified 104 capacity-relief periods.

8. KPI ANALYSIS

The final KPI analysis produced the following results:

KPI	Mean	Median	Minimum	Maximum
Total Children Under Care	6,232.77	6,633.50	2,002	11,762
Net Intake Pressure	-44.74	-11.50	-465	206
Care Load Volatility Index	1.25	0.82	0	11.60
Backlog Accumulation Rate	-45.94	-11.50	-402	112.71
Discharge Offset Ratio	167.76%	124.79%	4.55%	1,295.65%

The average total system load was approximately **6,233 children**, while the median was approximately **6,634 children**.

The maximum observed system inventory was **11,762 children**.

The mean net intake pressure was negative at **-44.74**, indicating that, across the analyzed reporting periods, the measured flow-balance indicator was more frequently negative than positive.

The care-load volatility index had a mean of **1.25** and a median of **0.82**, indicating variation in the rate of change of system load across reporting periods.

The discharge offset ratio was calculated only under the project's specified handling rule for periods where daily transfers met the minimum threshold.

9. SYSTEM LOAD THRESHOLD ANALYSIS

System-load thresholds were calculated using distributional percentiles.

Threshold	System Load
75th percentile	8,249.50
90th percentile	9,920.90
95th percentile	10,964.30

The final load classification contained:

Load Level	Number of Periods
Normal	540
Elevated	108
High	36
Very High	36

This classification indicates that 72 reporting periods were categorized as either High or Very High under the final system-load classification.

The classification provides a useful way to distinguish ordinary system conditions from periods where total care inventory was substantially above the historical distribution.

10. PEAK SYSTEM LOAD

The highest recorded system inventory occurred on:

December 20, 2023

At that time:

- **Total children under care:** 11,762
- **Children in CBP custody:** 246
- **Children in HHS care:** 11,516

The result demonstrates that the majority of the peak system inventory was located within HHS care rather than CBP custody.

This distinction is important because system-wide inventory alone does not indicate where operational resources are concentrated.

11. LOWEST SYSTEM LOAD

The lowest recorded system inventory occurred on:

August 24, 2025

At that time:

- **Total children under care:** 2,002
- **Children in CBP custody:** 21
- **Children in HHS care:** 1,981

The contrast between the peak and minimum observations demonstrates substantial variation in total system inventory over the analyzed period.

12. PROLONGED HIGH-LOAD WINDOWS

The analysis identified two major prolonged high-load windows.

Window 1

August 22, 2023 – October 19, 2023

- Duration: 41 reporting periods
- Peak load: 11,382 children
- Average load: 10,817.56 children

Window 2

November 21, 2023 – January 4, 2024

- Duration: 30 reporting periods
- Peak load: 11,762 children
- Average load: 10,975.10 children

These windows are particularly important from a capacity-management perspective because the high inventory persisted across multiple reporting periods rather than appearing as isolated single-day peaks.

Sustained high-load periods can have different operational implications from short-lived spikes because staffing, shelter capacity, medical services, and placement operations may need to remain elevated for an extended period.

13. CAPACITY RELIEF PERIODS

The refined analysis identified **104 capacity-relief periods**.

Among the strongest relief observations were periods in August 2025.

For example:

August 24, 2025

- Total system load: 2,002
- 7-period net intake average: approximately -5.29
- CBP custody: 21
- HHS care: 1,981

Several other strong relief observations occurred during August 2025, when total system inventory remained close to approximately 2,000 children.

These observations indicate a substantial reduction in system inventory compared with the high-load conditions observed during late 2023.

14. YEARLY CAPACITY ANALYSIS

The yearly analysis demonstrated a major change in system load across the three years.

Year	Average System Load	Peak System Load	Average Net Intake	Average HHS Care	Average CBP Care
2023	8,846.54	11,762	-130.96	8,646.07	200.47
2024	7,319.86	10,994	3.44	7,042.96	276.90
2025	2,575.74	6,471	-12.36	2,542.84	32.91

The average system load declined from **8,846.54 children in 2023** to **7,319.86 in 2024**, followed by a much larger decline to **2,575.74 in 2025**.

Average HHS care followed a similar pattern:

- 2023: 8,646.07
- 2024: 7,042.96
- 2025: 2,542.84

Average CBP care also decreased substantially by 2025.

These results indicate that the system operated at substantially lower observed inventory levels in 2025 than in 2023.

15. FORECASTING

A system-load forecast was generated as part of the project and exported as:

UAC_System_Load_Forecast.csv

The forecast was incorporated into the Streamlit dashboard alongside the historical system-load series.

The forecast should be interpreted as an analytical projection based on the available historical data and modeling approach used in the project.

It should not be interpreted as an official government forecast or as a guaranteed prediction of future UAC enrollment, custody, or care requirements.

Forecasting is nevertheless useful for scenario awareness because it allows historical patterns to be compared with projected system behavior.

16. DATA VISUALIZATION

The project incorporated visual analytics to improve interpretation of the operational data.

The primary visualizations included:

1. Total system load over time.
2. Seven-period and fourteen-period rolling averages.
3. CBP custody versus HHS care.
4. Net intake pressure.
5. Backlog accumulation.
6. Capacity-load classification.
7. High-load reporting periods.
8. Prolonged high-load windows.
9. Capacity-relief periods.
10. Yearly capacity comparison.
11. Historical system load versus forecast.

The final Streamlit dashboard provides interactive controls including date-range selection and time-granularity options.

17. STREAMLIT DASHBOARD

An interactive Streamlit application was developed to make the analytical results accessible through a web interface.

The dashboard contains:

- Executive KPI cards.
- System-load overview.
- CBP versus HHS comparison.
- Net intake and backlog analysis.
- Capacity classification.
- High-load period analysis.
- Prolonged stress windows.
- Relief-period analysis.
- Yearly capacity analysis.
- System-load forecast.

- Data download functionality.
- Methodology and KPI definitions.

The dashboard was implemented using Python, Pandas, Streamlit, and Plotly.

The live application is available at:

<https://uac-system-capacity-analytics-xrrkiouyv2djskoyp5dckr.streamlit.app/>

The project source code and analytical files are maintained in the GitHub repository:

<https://github.com/roshinisundaram4-code/UAC-System-Capacity-Analytics>

18. KEY FINDINGS

The major findings from the completed analysis are:

18.1 Substantial variation in system load

The system load ranged from **2,002 to 11,762 children**, demonstrating substantial variation across the analyzed period.

18.2 Late 2023 contained the most significant sustained high-load conditions

Two prolonged high-load windows were identified during 2023 and early 2024.

The longest identified window lasted **41 reporting periods**, while the second lasted **30 reporting periods**.

18.3 HHS care represented the majority of system inventory

During the peak system-load observation, **11,516 of 11,762 children** were in HHS care.

This indicates that HHS care inventory was the dominant contributor to total system load during the highest-load periods.

18.4 System inventory declined substantially by 2025

Average system load decreased from **8,846.54 in 2023** to **2,575.74 in 2025**.

The peak annual system load also decreased from **11,762 in 2023** to **6,471 in 2025**.

18.5 Relief conditions became more visible in 2025

The strongest relief observations occurred in August 2025, when system inventory was approximately 2,000 children.

18.6 Capacity monitoring should consider duration, not only peaks

A single high-load day and a 30- or 41-period high-load window have different operational implications.

Therefore, sustained-window analysis is important alongside daily threshold monitoring.

19. RECOMMENDATIONS

Based on the analytical findings, the following recommendations are proposed.

19.1 Establish continuous system-load monitoring

A centralized dashboard should monitor:

- Total children under care.
- CBP custody.
- HHS care.

- Transfers.
- Discharges.
- Net intake pressure.

This allows operational teams to identify changes in system conditions earlier.

19.2 Monitor prolonged high-load conditions

Capacity planning should not rely exclusively on daily maximum values.

Sustained high-load windows should be tracked because prolonged pressure can require continued staffing, shelter, medical, and placement resources.

19.3 Use percentile-based thresholds

The 75th, 90th, and 95th percentile thresholds provide data-driven reference points for identifying unusually high system inventory.

These thresholds can complement operational thresholds established by responsible agencies.

19.4 Integrate CBP and HHS inventory monitoring

Because total system load is distributed across different stages, monitoring only one component may hide pressure elsewhere.

Combined CBP and HHS monitoring provides a more complete picture.

19.5 Use forecasting for preparedness

Forecast outputs can support scenario planning and resource-preparation exercises.

Forecasts should be updated as new operational data become available and should be interpreted together with current conditions rather than used as standalone predictions.

19.6 Maintain transparent analytical definitions

Metrics such as net intake pressure, backlog accumulation, volatility, and discharge offset should have clearly documented definitions.

This is important for ensuring consistent interpretation across users.

20. LIMITATIONS

Several limitations should be considered.

First, the analysis is based on the supplied operational dataset and therefore reflects only the variables available in that dataset.

Second, the project measures operational flows and inventories rather than directly measuring healthcare quality, individual patient outcomes, clinical outcomes, or wellbeing.

Third, a high system load does not automatically demonstrate inadequate care capacity. It indicates increased inventory and potential operational pressure that requires contextual interpretation.

Fourth, negative or positive flow-balance indicators should not be interpreted as direct causal explanations for changes in system inventory.

Fifth, the forecast represents an analytical projection and is subject to uncertainty. Future operational conditions may differ from historical patterns.

Finally, the findings should be used as decision-support information rather than as a substitute for operational judgment or official policy guidance.

21. CONCLUSION

This project developed a structured analytical framework for monitoring system capacity and care load within the UAC care pipeline.

Using 720 reporting periods from January 2023 through December 2025, the analysis transformed operational records into system-load metrics, flow indicators, rolling averages, capacity classifications, stress windows, relief periods, yearly comparisons, and a forecast.

The analysis identified a peak system inventory of **11,762 children on December 20, 2023** and demonstrated that the majority of that inventory was in HHS care.

Two prolonged high-load windows were identified during late 2023 and early 2024. In contrast, 2025 showed substantially lower average and peak system-load levels, with several pronounced relief periods.

The resulting Streamlit dashboard provides an interactive method for exploring these patterns and supports operational situational awareness through KPI cards, time-series visualization, threshold analysis, stress-period identification, and forecasting.

Overall, the project demonstrates how data science and healthcare analytics can transform operational data into a structured capacity-monitoring framework. Such an approach can support more proactive planning, improve visibility into system conditions, and provide a foundation for ongoing analysis as new data become available.

22. PROJECT DELIVERABLES

The completed project includes:

- Data preparation and validation.
- Capacity metric generation.
- KPI analysis.
- System-load threshold classification.
- High-load period identification.
- Prolonged stress-window detection.
- Capacity-relief analysis.
- Yearly capacity analysis.
- System-load forecasting.
- Executive findings.
- Complete analytical dataset.
- KPI summary dataset.
- Yearly capacity summary.
- High-load period dataset.
- Relief-period dataset.
- Forecast dataset.
- Excel analytical workbook.
- Streamlit interactive dashboard.
- GitHub project repository.

Project Links

GitHub

<https://github.com/roshinisundaram4-code/UAC-System-Capacity-Analytics>

Repository:

Live

Streamlit

Dashboard:

<https://uac-system-capacity-analytics-xrrkiyuv2djskoyp5dckr.streamlit.app/>